

Labwork 8: Scatter

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1 Introduction

This report presents two scatter kernels on GPU: `RGB2HSV()` converts an RGB image to HSV, and `HSV2RGB()` converts it back.

2 Method

The RGB image is stored as an **Array of Structs (AoS)**: one array `img[y, x, c]` where the 3 channels of a pixel sit next to each other. The HSV result is stored as a **Struct of Arrays (SoA)**: three separate 2D arrays `H[y, x]`, `S[y, x]`, `V[y, x]`.

Both kernels use a 16×16 thread block, one thread per pixel. Test image: a 3840×2160 RGB image.

3 RGB to HSV

Each thread reads its pixel, scales R, G, B to $[0, 1]$, and computes max , min and $\Delta = max - min$:

$$H = \begin{cases} 0^\circ & \Delta = 0 \\ 60^\circ \times (\frac{G-B}{\Delta} \bmod 6) & max = R \\ 60^\circ \times (\frac{B-R}{\Delta} + 2) & max = G \\ 60^\circ \times (\frac{R-G}{\Delta} + 4) & max = B \end{cases} \quad S = \begin{cases} 0 & max = 0 \\ \frac{\Delta}{max} & max \neq 0 \end{cases} \quad V = max \quad (1)$$

The “mod 6” is done by adding 360° when H is negative. The thread then **scatters** the three values into the three output arrays:

```
H[y, x] = h
S[y, x] = s
V[y, x] = mx
```

4 HSV to RGB

Each thread reads H, S, V from the three arrays and computes

$$d = H/60, \quad hi = \lfloor d \rfloor \bmod 6, \quad f = d - \lfloor d \rfloor \quad (2)$$

$$l = V(1 - S), \quad m = V(1 - fS), \quad n = V(1 - (1 - f)S) \quad (3)$$

(R, G, B) is picked from (V, n, l) , (m, V, l) , (l, V, n) , (l, m, V) , (n, l, V) , (V, l, m) depending on hi . It is scaled back to $[0, 255]$ (rounded) and written into the single RGB array.

5 Results

The image was converted to HSV and back to RGB, and the output was compared with the input. The maximum per-channel difference is **0**, so the round trip is exact.

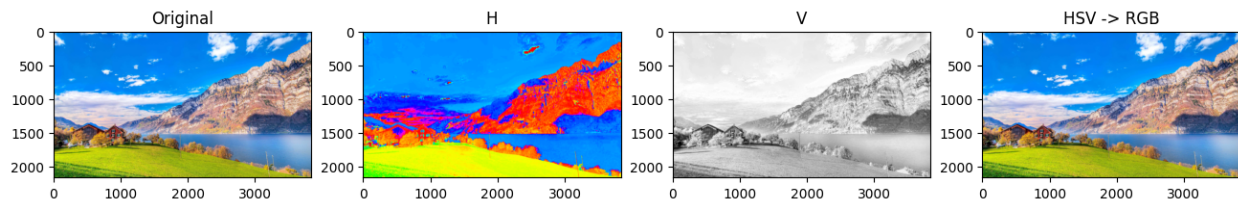


Figure 1: Original, H channel, V channel, and image converted back to RGB